

Muhammad Akif Naveed

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ACADEMIC OBJECTIVE

Software Engineering student at COMSATS University Islamabad with a CGPA of 3.62/4.0, seeking the COMSATS-METU exchange opportunity to gain international academic exposure in AI and intelligent systems. Combines strong coursework with independent AI engineering exploration, hackathon participation, research assistance, and leadership in technical communities. Particularly interested in AI-assisted systems, intelligent automation, and applied deep learning for real-world problem solving and research-oriented development.

EDUCATION

COMSATS University Islamabad, Wah Campus

2024 – 2028

Bachelor of Science in Software Engineering | CGPA: 3.62 / 4.0 | 4th Semester

Relevant Coursework: Programming Fundamentals, Discrete Structures, OOP, Database Systems, Computer Networks, Data Structures & Algorithms, Software Engineering, Digital Logic Design

Academic Performance: Sem 1 GPA 3.59 | Sem 2 GPA 3.75 | Sem 3 GPA 3.68 | Sem 4 CGPA 3.62

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL, HTML/CSS

AI / ML: Machine Learning, Deep Learning, Computer Vision, NLP, TensorFlow, PyTorch, Scikit-learn, OpenCV, Pandas, NumPy, Streamlit

Systems & Tools: Git/GitHub, VS Code, Jupyter, Google Colab, Antigravity, MySQL, SQLite, Supabase, Qt Framework, Tkinter, Java Swing

APIs & Deployment: Groq API, Hugging Face, Streamlit Cloud, Vercel, FAST APIs

ACADEMIC & TECHNICAL PROJECTS

NOVA AI — Voice-Controlled Intelligent Personal Assistant [\[GitHub\]](#)

Semester 4 · Python, NLP, Groq API, Speech Recognition, Automation · AI Engineering

- Developed an NLP-based voice assistant integrating intent recognition, real-time automation, and backend memory management for contextual responses.
- Integrated conversational AI workflows and gesture-based interaction mechanisms to explore intelligent human-computer interaction within voice-controlled assistant systems, applying concepts from AI Engineering coursework independently.

Evidentia.pk — AI-Powered Digital Evidence Platform [\[GitHub\]](#) [\[Live deployed app\]](#)

HEC GenAI Cohort 2 Hackathon · Python, Streamlit, Groq API, Supabase · Legal-Tech AI

- Built an AI-driven platform for secure evidence sealing with timestamping, image-to-text inference, and AI-assisted evidence interpretation within a 72-hour development window.
- Integrated Supabase for backend data management and led system integration across AI pipeline, frontend, and deployment — project successfully demonstrated at the competition.

Waris Nama AI — AI-Assisted Heritage & Legal Guidance Platform [\[GitHub\]](#) [\[Live Deployed App\]](#)

HackData V1 · Python, AI Integration, Web Frontend · 4th Position · Legal-Inheritance-Tech AI

- Collaboratively developed an AI-assisted platform focused on simplifying inheritance-related guidance and workflow accessibility through intelligent user interaction and automated information assistance.
- Integrated AI-powered workflows and PDF Generation components within a competitive hackathon environment, contributing to rapid prototyping, feature implementation that secured 4th position overall

CureOne — Unified Healthcare Management System [\[GitHub\]](#)

Semester 3 · Java, Java Swing, MySQL · OOP & Database Systems

- Developed a modular desktop healthcare application applying OOP design patterns, relational database integration (MySQL), and GUI development — encompassing pharmacy, appointment, and patient record modules.

Network-Controlled Custom CMD Panel via TCP Sockets [\[GitHub\]](#)

Semester 3 · Python, TCP Sockets, Tkinter, Multi-threading · Computer Networks

- Implemented client-side architecture for a multi-client TCP command execution system with a GUI panel, JSON-based message framing, and multi-threaded server handling — demonstrating client-server networking fundamentals.

CDSIAS — Comprehensive Data Structure Implementation & Analysis System [\[GitHub\]](#)

Semester 4 · C++, Qt Framework, SQLite · Data Structures & Algorithm Analysis

- Engineering a modular C++ desktop application implementing all fundamental data structures (Trees, Graphs, Hashing, Sorting) from scratch without STL, with real-time performance analytics stored in SQLite.

AlgoVision — AI-Powered Data Structures Learning & Analysis System [\[GitHub\]](#)

Semester 4 · Python, AI Integration, Web Frontend · Software Requirements Engineering

- Proposed and designed an intelligent educational platform combining AI-assisted learning, algorithm visualization, and automated performance feedback for data structure education.

Face Recognition-Based Student Attendance System [\[GitHub\]](#)

AI Engineering Program Final Project · Python, OpenCV, Face Recognition, SQLite · Computer Vision

- Developed an automated attendance management system using real-time face detection and recognition via OpenCV, built as the capstone project of PNY AI Engineering program.
- Implemented a face encoding pipeline for student attendance with liveness, anti-spoofing, live camera based recognition, and automated attendance logging into a structured SQLite database along with separate dashboards for admin and students.

Brain Tumor MRI Classification — Ensemble Deep Learning

Ongoing Research Assistance · CNN, ResNet50, EfficientNet, Transfer Learning, CBAM Attention

- Assisting in research on multi-class brain tumor classification from MRI scans using an ensemble of CNN, ResNet50, and EfficientNet architectures with Grad-CAM visualization for interpretability.
- Contributing to model experimentation, accuracy optimization, and validation stability analysis for the hybrid ensemble framework.

RESEARCH & TRAINING

Research Assistant — Brain Tumor Classification (Deep Learning)

2025 – Present

Assisting faculty in applying ensemble deep learning (CNN, ResNet50, EfficientNet) with attention mechanisms on MRI datasets; focus on performance optimization and model evaluation.

AI Engineering Program — PNY Trainings, Lahore

6 Months · Completed

Completed a practical AI engineering program covering ML workflows, deep learning concepts, intelligent application development, API integration, and deployment pipelines — extending significantly beyond university coursework.

HEC GenAI Cohort 2 Training Program

6 Weeks · Completed

Completed HEC's Generative AI cohort covering LLM integration, AI-assisted workflow design, and collaborative AI system development; culminated in the Evidentia.pk hackathon project.

HACKATHONS & ACHIEVEMENTS

- 4th Position, HackData V1 Hackathon — Waris Nama AI project (rapid AI prototyping and team collaboration under competition conditions).
- HEC GenAI Cohort 2 Hackathon — Developed and deployed Evidentia.pk, an AI-powered legal-tech platform, within a 72-hour competitive window.
- Best Student Award — Data Science & AI Club, GDG On Campus, COMSATS Wah Chapter (Fall 2025).
- 14th Position — Speed Coding Competition, Tech Fest 2026, COMSATS University Islamabad.

LEADERSHIP & COMMUNITY INVOLVEMENT

Campus Ambassador — CodeX 3.0, NUML University Mianwali

2025

Campus Ambassador — RoboFiesta, Hi-Tech University Islamabad

2025

Event Management — Visio Spark 2025, COMSATS University (20+ universities)

2025

Active Member — Google Developers Group On Campus, CUI Wah Chapter

2025–Present

Conducts peer mentorship sessions guiding juniors on AI learning pathways, hackathon participation, and technical career development.

CERTIFICATIONS

AI Engineering Program — PNY Trainings, Lahore

6 Months · Completed

Focus: Machine learning fundamentals, deep learning, intelligent application development, API integration, deployment workflows.

Capstone: Face Recognition-Based Student Attendance System

NCEAC - HEC GenAI Cohort 2 — Generative AI Training Program

6 Weeks · Completed

Higher Education Commission of Pakistan (Virtual)

Focus: Large language models, prompt engineering, AI-assisted workflow design, collaborative AI development.